7. ELLIPTIC CURVE CRYPTOGRAPHY (ECC)

AIM

To implement Elliptic Curve Cryptography (ECC) using Python.

Algorithm

1. Start.
2. Generate ECC private and public keys.
3. Enter the plaintext.
4. Encrypt the plaintext.
5. Decrypt the ciphertext.
6. Display the original plaintext.
7. Stop.

Python Code

```
from Crypto.PublicKey import ECC
from Crypto.Hash import SHA256
message = input("Enter the Plain Text: ")
key = ECC.generate(curve='P-256')
public_key = key.public_key()
hash_object = SHA256.new(message.encode())
print("\nECC Public Key:")
print(public_key.export_key(format='PEM'))
print("\nECC Private Key:")
print(key.export_key(format='PEM'))
print("\nSHA-256 Hash of Message:")
print(hash_object.hexdigest())
```

Output

```
Enter the Plain Text: hello world

ECC Public Key:
-----BEGIN PUBLIC KEY-----
MFkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDQgAErCcm6oepQWLTyBzErVdXpU+yHaP2
8mXyxcDIsdtTXcz5BL7nSK+uOgObkuyIGQioxKAMVouX4YGxAk3ckmK9eg==
-----END PUBLIC KEY-----

ECC Private Key:
-----BEGIN PRIVATE KEY-----
MIGHAgEAMBMGByqGSM49AgEGCCqGSM49AwEHBG0wawIBAQQgZ8XU1OCIpsODt+eO
UbEUt1pwbULWTHnxyg9vlVdJkNuhRANCAASsJybqh6lBYtPIHMStV1elT7Ido/by
ZfLFwMix21NdzPkEvudIr646A5uS7IgZCKjEoAxWi5fhgbECTdySYr16
-----END PRIVATE KEY-----

SHA-256 Hash of Message:
b94d27b9934d3e08a52e52d7da7dabfac484efe37a5380ee9088f7ace2efcde9
```

The result:

The above code is executed and the output is obtained successfully.